

IMPORTANT STUDENT INSTRUCTIONS

Ensure your **Student Number** and **date of birth** are entered in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers)

2	8	3	9	7	6	8
N	N	N	N	N	N	N

Date of Birth (DD/MM/YYYY)

0	9	/	0	3	/	2	0	0	4
D	D		M	M		Y	Y	Y	Y

COMPULSORY Q2.

Do not mark any other boxes.

☐ DO NOT USE

☒ Q2

☐ DO NOT USE

☐ DO NOT USE

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☐ DO NOT USE

You may use both sides but only answer **ONE** question per sheet

Q2

(a).

$$X(z) = Z(x(n)) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} \quad \checkmark \quad +1$$

the difference equation of the trapezoid integration:

$$y_n - y_{n-1} = \frac{(x_n + x_{n-1})T}{2} \quad \checkmark$$

$$\Rightarrow Y(z) - Y(z) \cdot z^{-1} = \frac{T}{2} X(z) + \frac{T}{2} X(z) \cdot z^{-1} \quad (x_0=0, y_0=0)$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{\frac{T}{2}(1+z^{-1})}{1-z^{-1}}$$

$$\text{or } \frac{X(z)}{Y(z)} = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$$

How? x

+3

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(b).

(i) Aliasing happens when the sample frequency is smaller than the twice bandwidth of the sampled signal. This will lead to the overlap of adjacent frequency. $\checkmark \quad +3$

Therefore, this can decrease the frequency measurement accuracy.

(ii).

$f_s > 2B$ EXPLAIN! x +1

(c) For backward rectangular rule:

$$\frac{y_n - y_{n-1}}{T} = x(n) \quad \checkmark$$

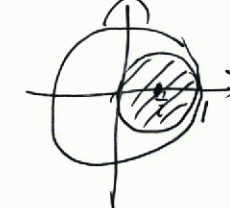
$$\Rightarrow Y(z) - Y(z) \cdot z^{-1} = X(z) \cdot T$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{T}{1-z^{-1}} = \frac{zT}{z-1} \rightarrow s$$

$$\Rightarrow z = \frac{s}{s-T} = \frac{1}{2} + \frac{s}{s-T} - \frac{1}{2}$$

$$= \frac{1}{2} + \frac{s+T}{2(s-T)}$$

$$\Rightarrow |z - \frac{1}{2}| = \left| \frac{s+T}{2(s-T)} \right| = \frac{1}{2}$$



Therefore, it maps poles into a circle with $r = \frac{1}{2}$ at $\frac{1}{2}$. It can preserve the stability.

no!! x

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+3

Q2 Continued

You may use both sides but only answer **ONE** question per sheet

(d).
(i) Because in ~~frequency~~ ^{time} domain, it is a convolution between $\text{sinc}(\frac{\pi(t-kT)}{T})$, it is a multiplication between $R(k)$ and a rectangular window ~~with length~~ in the ~~time~~ ^{frequency} domain. no X

To ensure exact reconstruction, we should ensure the bandwidth of ~~$R(k)$~~ ^{$r(t)$} smaller than the window length. i.e. $B < \frac{\pi}{T}$. no X

(ii) Because in a realistic scenario, it is impossible to perform $\sum_{k=-\infty}^{+\infty}$, there should be a finite range. ✓ + 2

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